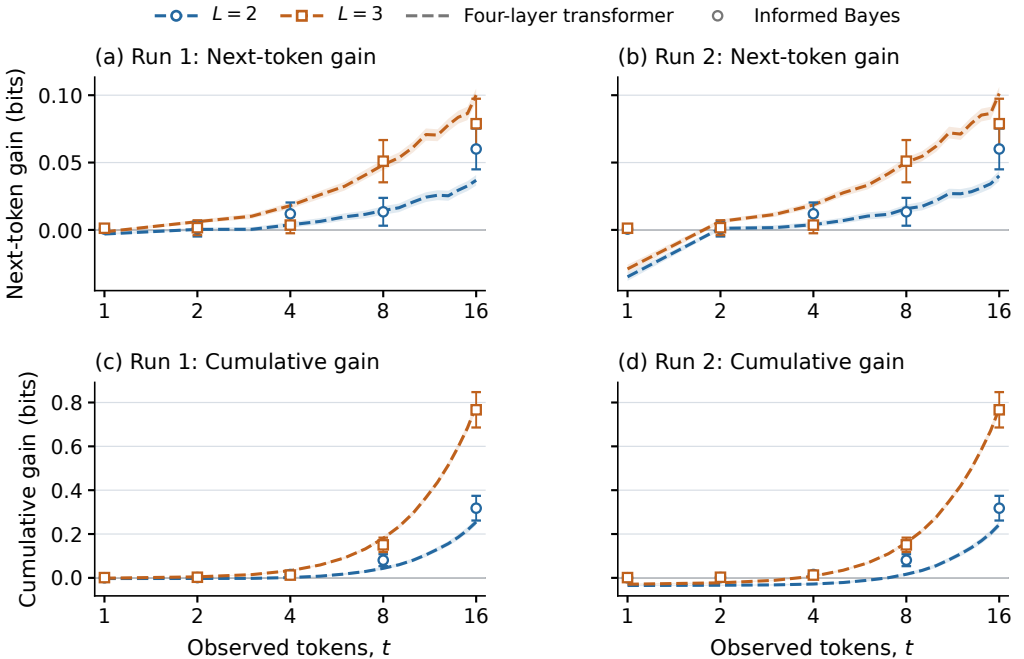
$L = 2$
$L = 3$
Four-layer transformer
Informed Bayes
(a) Run 1: Next-token gain
(b) Run 2: Next-token gain
(c) Run 1: Cumulative gain
(d) Run 2: Cumulative gain
Next-token gain (bits)
Cumulative gain (bits)
0.10
0.05
0.00
0.8
0.6
0.4
0.2
0.0
1
2
4
8
16
Observed tokens, t